Sungsu Kim

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EDUCATION

New York University Abu Dhabi, Abu Dhabi, UAE — Expected Graduation Date (May 2027)
Bachelor of Science, Computer Engineering | GPA: 3.99 / 4.00

- Study Away: NYU Tandon School of Engineering, New York, USA | Sep 2025 – May 2026 | GPA: 4.00 / 4.00
- Relevant Coursework: Robot Vision, Artificial Intelligence, Machine Learning, Control Systems Engineering, Signals and Systems, Introduction to Embedded Systems Design, Design and Analysis of Algorithms

Gems American Academy, Abu Dhabi, UAE — Graduation Date (May 2021)
IB Diploma: 44 / 45 | Graduated with the highest honor

RESEARCH INTERESTS

- High-level task planning and motion planning for manipulation and navigation
- Perception and spatial understanding with point clouds, SLAM, and 3D Gaussian Splatting (3DGS)
- Sim-to-real transfer and deployment with a focus on safety and human–robot interaction

PUBLICATIONS AND MANUSCRIPTS

[1] **S. Kim**, V. Bhat, P. Krishnamurthy, R. Karri, and F. Khorrami, "CLARO: Clutter-Aware Robot Orchestrator for Open-Vocabulary Grasping Using VLM Agents," *IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*, 2027. **(Accepted, to appear)**

[2] V. Bhat, **S. Kim**, V. Blukis, G. Heinrich, P. Krishnamurthy, R. Karri, S. Birchfield, F. Khorrami, and J. Tremblay, "BOP-ASK: Object-Interaction Reasoning for Vision-Language Models," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.

RESEARCH EXPERIENCE

AI4CE Lab, New York, USA — Jun 2026 – Present
Undergraduate Research Assistant | Prof. Chen Feng

- Developed long-horizon (300 m) outdoor **humanoid navigation** for a Unitree G1, including crosswalks with dynamic pedestrians and vehicles.
- Implemented a **safety layer** based on time-to-collision (TTC) and control barrier function (CBF) for dynamic obstacle avoidance in simulation and on G1.
- Evaluated GuideNav and fine-tuned Uni-NaVid **vision-language policies** in Isaac Sim and high-fidelity 3D Gaussian Splatting (3DGS) environments.
- Fine-tuned the low-level control policy and evaluated it over 590+ episodes, benchmarking success rate, SPL, navigation error, and path length.

Control/Robotics Research Laboratory, New York, USA — Oct 2025 – Jun 2026
Undergraduate Research Assistant | Prof. Farshad Khorrami

- Curated a 150K-image dataset for **VLM's special and object-interaction reasoning** over pose, grasp, trajectory, and spatial relations; fine-tuned NVILA and Qwen on it [2].
- Designed a **VLM orchestrator** for **high level planning** in occlusion-aware open-vocabulary grasping and benchmarked grasping methods in simulation with high-quality Blender rendering [1].
- Conducted 240+ **real-robot experiments** across 16 scenes with 40 distinct objects on a **Franka arm**; adapted top-down view to an oblique camera setup using point-cloud projection and morphological post-processing.

Center for Artificial Intelligence and Robotics, Abu Dhabi, UAE — Feb 2025 – Sep 2025
Undergraduate Research Assistant | Prof. Anthony Tzes

- Developed **audio-enhanced semantic SLAM** by integrating auditory cues into RTAB-Map for loop closure and deployed the system on a Diablo wheeled-bipedal robot for real-world experiments.
- Fine-tuned Google YAMNet for domestic sound classes with a Bi-LSTM structure (transfer learning on FSD50K dataset and low-pass-filter augmentation); improved classification accuracy from 42% to 79%.
- Localized sound sources in 3D using an SV600 acoustic imager and least-squares estimation, achieving a localization error of 0.063 m.

Applied Interactive Multimedia Laboratory, Abu Dhabi, UAE Jun 2024 – Jul 2024
Undergraduate Research Assistant | Prof. Mohamad Eid

- Built an **XR environment** in Unity on Meta Quest 3 that delivers immersive visual, auditory, and haptic stimuli to assist in regulating emotion, stress, anxiety, and depression.

PROJECTS

Senior Capstone Design Project, Abu Dhabi, UAE Aug 2026 – Present

- Investigated **object navigation** when several instances of the target object are present in domestic and industrial environments.
- Developed an **embodiment-agnostic** modular design for the TITA robot and Boston Dynamics Spot.
- Planned to combine a scene graph with a functional graph and 3DGS for spatial understanding.

Visual Navigation in Unseen Mazes, New York, USA Feb 2026 – May 2026

- Developed a fully autonomous **visual navigation system** for unseen mazes from exploration data; placed **1st in a course challenge**, achieving a navigation time of 9.05 s (51.7% lower than the 2nd) and a position error of 0.051 m.
- Built offline maps from odometry and wall detections using Manhattan optimization, collision correction, and false-wall removal; process map with open, uncertain, and unexplored occupancy cells.
- Localized the initial robot pose and the goal position using SIFT features, VLAD image retrieval, and essential-matrix refinement; planned low-turn routes with direction-aware A* and sparse waypoints for autonomous execution and goal check-in.

Drone Simulation with RotorS, Seoul, South Korea May 2024

- Developed a drone simulation on Ubuntu using **ROS** and **Gazebo** with the ETH Zurich RotorS odometry package and designed the URDF defining the drone's properties and configuration.
- Implemented ROS nodes, servers, and services to take off, navigate through waypoints, hover while recording with the onboard camera, and land.

AWARDS, GRANTS, AND SELECTED WRITING

Summer Research Grant, NYU Abu Dhabi – $4000 Apr 2026
Exit 11: A journal of the First-Year Writing Seminars *2020-2021 & 2021-2022*, NYU Abu Dhabi, Jun 2024
issue 5 & 6, pg. 363-371
U.S. Army Commendation Medal (ARCOM), U.S. Army Apr 2024

LEADERSHIP AND SERVICE

Engineers for Social Impact, Kabwe, Zambia Mar 2025
NYU Abu Dhabi and Habitat for Humanity International

- Worked with local builders and Habitat for Humanity to construct a home in the Makululu community.
- Conducted community-based fieldwork examining how technology, communication, cooperation, and trust shape everyday life and community development.

Army Service, U.S. Army Garrison, Pyeongtaek, South Korea Oct 2022 - Apr 2024
Supply Specialist | Korean Augmentation to the US Army (KATUSA)

- Managed **$41 million in property** across 35 sub-hand receipts; coordinated 50+ equipment turn-ins and in and out processing for 400+ soldiers.
- Reconciled 1,000+ bills of materials in 30 days; rectified $15,000 in shortages before inventories and closed with two statements of charges worth less than $2,000; recognized as **Soldier of the Battalion** for these contributions.
- Served as a soldier counselor and mentored 33 KATUSA soldiers in equipment requisition.

SKILLS

Languages: English (Fluent, TOEFL 110), Korean (Native)
Software: Python, C++, MATLAB, ROS/ROS 2, Unity, TensorFlow, PyTorch, VHDL, Cadence Virtuoso, FPGA, Arduino, Isaac Sim, Gazebo, Blender, MuJoCo